

Editorial Perspective: The need for new guidelines to improve the reporting of trials in child and adolescent mental health

Frances Gardner,¹ Evan Mayo-Wilson,² Paul Montgomery,¹ Sally Hopewell,³ Geraldine Macdonald,⁴ David Moher,⁵ and Sean Grant¹

¹Centre for Evidence-Based Intervention, University of Oxford, Oxford, UK; ²Centre for Outcomes Research and Effectiveness, Research Department of Clinical, Educational & Health Psychology, University College London, London, UK; ³Centre for Statistics in Medicine, University of Oxford, Oxford, UK; ⁴Institute of Child Care Research, Queen's University Belfast, UK; ⁵Clinical Epidemiology Program, Ottawa Hospital Research Institute Centre for Practice-Changing Research (CPCR), The Ottawa Hospital - General Campus, Ottawa, ON, Canada

We are pleased to announce the development of CONSORT-SPI: a new reporting guideline for randomised controlled trials (RCTs) of social and psychological interventions. We invite the readership of *Journal of Child Psychology and Psychiatry* (JCPP) to participate in this project by visiting our website (<http://tinyurl.com/CONSORT-study>).

Well-designed and conducted randomised trials are essential for improving services for children and young people. Not only can they help us to establish 'what works' in child and adolescent mental health (Maughan, 2013), but they can also help us to understand how interventions work, and to identify ways in which individual characteristics affect susceptibility to intervention effects. The JCPP has led the way in promoting a high-quality evidence-base for interventions in this field. Over a decade ago, Harrington, Cartwright-Hatton, and Stein (2002) pointed to the importance of 'well-designed and unambiguously reported RCTs [...to...] provide the best possible evidence about effectiveness of interventions,' and drew attention to the many special considerations entailed in designing and interpreting RCTs in the field of child mental health. For example, they describe how interventions in this field are complex, involving multiple systems and professions. Moreover, interventions operate against a backdrop of developmental change, and multiple, interacting levels of social influence on the child. Measurement of outcomes often requires multi-method assessment of child well-being, psychopathology, and co-morbidities, as well as family functioning. Thus, given the complexity of these interventions and their outcomes, they are difficult to evaluate.

To understand intervention effects and to keep services up to date, researchers, clinicians, commissioners, policymakers, journalists, and consumers rely on research reports of RCTs in scientific journals. Such reports should explain the methods, including the design, delivery, uptake, and context of interventions, and subsequent results. Accurate, complete, and transparent reporting is essential for

readers to make best use of new evidence, to achieve returns on research investment, to meet ethical obligations to research participants and consumers of interventions, and to minimise harm to children and waste of resources. Despite this need, RCTs are often insufficiently reported to understand internal validity (bias) and external validity (the extent to which a study's results are applicable in other settings or populations). Literature reviews indicate that previous efforts to improve reporting in psychosocial disciplines can be useful and influential yet fall short for RCTs compared to medical disciplines (Grant, Mayo-Wilson, Melendez-Torres, & Montgomery, 2013; Stinson, McGrath, & Yamada, 2003). Poorly reported RCTs of child-focused interventions lead to shortcomings in the reliability and utility of this evidence for improving child health. For interventions in child psychology and psychiatry to rely on the evidence provided by RCTs, the quality of reporting must be addressed.

Reporting guidelines help us use scarce resources efficiently and ethically. They aim to improve the quality of reports by listing the minimum information that is required to understand how studies were conducted. Reporting guidelines do not prescribe research conduct, but facilitate the writing and appraisal of reports. When reports of RCTs cannot be used to inform decision-making, resources are wasted. Furthermore, research participants contribute their time and put themselves at risk of harm to generate information that will help other people, and researchers should disseminate that information effectively. Practitioners learn to deliver interventions by reading reports with sufficient detail to allow replication; high quality reports thus facilitate high quality rollout and help practitioners inform users about the risks and benefits of different interventions.

The Consolidated Standards of Reporting Trials (CONSORT) Statement is an evidence-based guideline that includes a checklist of 25 items, a flow diagram, and examples of good reporting (Moher et al., 2010). CONSORT has improved the reporting

of thousands of medical experiments and has been endorsed by over 600 journals. Although Harrington et al. (2002) called for wider use of CONSORT standards to improve the quality of reporting RCTs in child mental health, arguably the field has not taken up this challenge. Researchers and journal editors in the social and behavioural sciences are generally aware of CONSORT but often object that it is not appropriate for social and psychological interventions. As a result, uptake of CONSORT guidelines in these disciplines is low – neither psychology nor specialty areas within psychology have extensively adopted CONSORT guidelines. Items within these guidelines may not sufficiently cover unique reporting needs for psychosocial interventions, such as those found in the child mental health field. Moreover, while social and behavioural scientists have developed other reporting guidelines, they have not been developed and disseminated using optimal methods (Grant et al., 2013). With increasing reliance on RCTs in policy-making and practice in child and adolescent mental health (Hartling et al., 2012), and greater public attention drawn to the need to improve the conduct and reporting of intervention trials (Moher et al., 2010), it is very timely to extend CONSORT guidelines to specifically address the important features of social and psychological interventions, such as those in child and adolescent mental health.

We invite stakeholders in the field of child psychology and psychiatry to participate in the development of CONSORT-SPI: an extension for social and psychological interventions. This guideline will incorporate lessons from previous extensions (Grant et al., 2013) and other reporting guidelines to aid the critical appraisal, replication, and uptake of this research. It will target several aspects of internal validity that are especially important for these trials in this field (Harrington et al., 2002), such as randomisation techniques, the use of appropriate outcome measures, and blinding of researchers to intervention condition(s) (Sonuga-Barke et al., 2013). It will also target several areas of external validity such as intervention theory of change, key components of interventions, representativeness of the sample and setting, and implementation issues such as delivery, participant uptake, and contextual influences. When testing interventions with and for children, these aspects relate to issues such as the recruitment of appropriately aged children, informed consent and assent, age-appropriate administration of interventions, and the validity of outcome measures for child populations (Saint-Raymond et al., 2010). We hope that the involvement of stakeholders within child and adolescent mental health research will help this guideline to include important standards for this field.

The project is being led by an international collaboration of researchers, methodologists, guideline developers, funders, service providers, and

journal editors. We will use consensus techniques to reduce bias and to promote widespread guideline uptake (Grant, Montgomery, & Mayo-Wilson, 2012). Following rigorous reviews of existing guidelines and current reporting quality, we will conduct a Delphi process to identify a prioritised list of reporting items to consider for the extension. Following the Delphi process, we will host a consensus meeting to review the findings and to generate a list of minimal reporting standards, mirroring the development of previous CONSORT guidelines.

CONSORT-SPI will help investigators in child psychology and psychiatry to improve the quality of reporting RCTs, thereby enhancing the scientific knowledge of child and adolescent services and facilitating the use of scientific evidence in policy and practice for the children and their families that this field aims to serve (Stinson et al., 2003). The success of this project depends on widespread involvement and agreement among key international stakeholders in research, policy, and practice. Many journal editors have already agreed to participate, and we hope other researchers and stakeholders will volunteer their time and expertise. The *JCPP* has a long-standing commitment to rigorous reporting, as evidenced by its requirement for authors to conform to reporting guidelines such as the CONSORT Statement and to register RCTs with a trial registry. We invite stakeholders from psychology, psychiatry, and related disciplines to join this important effort and participate in guideline development.

Supporting information

Additional Supporting Information may be found in the online version of this article:

Data S1. The CONSORT-SPI (Social and Psychological Interventions) International Advisory Group

Acknowledgement

This project is funded by the UK Economic and Social Research Council (ES/K00087X/1). We thank the Centre for Evidence Based Intervention (Oxford University), the Centre for Outcomes Research and Effectiveness (University College London), and the National Collaborating Centre for Mental Health (NCCMH) for their support. S.G. is supported by a linked Clarendon Fund-Green Templeton College Annual Fund Scholarship to support his doctoral studies and research. D.M. is supported by a University Research Chair. Details of The CONSORT-SPI (Social and Psychological Interventions) International Advisory Group is provided as online supporting material

References

- Grant, S.P., Mayo-Wilson, E., Melendez-Torres, G.J., & Montgomery, P. (2013). The reporting quality of social and psychological intervention trials: a systematic review of reporting guidelines and trial publications. *PLoS One*, 8, e65442. doi: 10.1371/journal.pone.0065442

- Grant, S., Montgomery, P., & Mayo-Wilson, E. (2012). Development of a CONSORT extension for interventions in public health and related disciplines. *The Lancet*, 380(Supp. 3). S14. Available at: <http://www.thelancet.com/abstracts/public-health-science-in-the-uk>.
- Harrington, R.C., Cartwright-Hatton, S., & Stein, A. (2002). Annotation: Randomised trials. *Journal of Child Psychology and Psychiatry*, 43, 695–704.
- Hartling, L., Wittmeier, K.D.M., Caldwell, P., van der Lee, H., Klassen, T.P., Craig, J.C., ... & for the StaR Child Health Group. (2012). StaR child health: Developing evidence-based guidance for the design, conduct, and reporting of pediatric trials. *Pediatrics*, 129, S112.
- Maughan, B. (2013). Editorial: "Better by design" – why randomized controlled trials are the building blocks of evidence-based practice. *Journal of Child Psychology and Psychiatry*, 54, 225–226.
- Moher, D., Hopewell, S., Schultz, K.F., Montori, V., Gøtzsche, P.C., Devereaux, P.J., ... & Altman, D.G. (2010). CONSORT 2010 explanation and elaboration: Updated guidelines for reporting parallel group randomised trials. *BMJ*, 340, c869.
- Saint-Raymond, A., Hill, S., Martines, J., Bahl, R., Fontaine, O., & Bero, L. (2010). CONSORT 2010. *The Lancet*, 376, 229.
- Sonuga-Barke, E.J.S., Brandeis, D., Cortese, S., Daley, D., Ferrin, M., Holtmann, M., ... & European ADHD Guidelines Group. (2013). Nonpharmacological interventions for ADHD: Systematic review and meta-analyses of randomized controlled trials of dietary and psychological treatments. *American Journal of Psychiatry*, 170, 275–289.
- Stinson, J.N., McGrath, P.J., & Yamada, J.T. (2003). Clinical trials in the Journal of Pediatric Psychology: Applying the CONSORT statement. *Journal of Pediatric Psychology*, 28, 159–167.

Accepted for publication: 15 May 2013

Erratum

The spelling of the surname of the first author should be corrected *from Karatoreos to Karatsoreos* on the title page and running headers for this article, both in print and in online versions. The correspondence details for the first author in the published version are correct.

Reference

- Karatoreos, I.N., & McEwen, B.S. (2013). Annual Research Review: The neurobiology and physiology of resilience and adaptation across the life course. *Journal of Child Psychology and Psychiatry*, 54, 337–347. doi:10.1111/jcpp.12054